CS 135: Computational Semantics

Fall 2026 — Course Plan (Revision 3): From Logic to Geometry

Course Description

This course studies the computational treatment of meaning in natural language, following a single question across two traditions: what structure does meaning composition require? The first unit develops the logical answer – the lambda calculus, semantic types, models, and a working interpreted fragment of English – with quantification and scope as the unit's capstone phenomena. The second unit develops the distributional answer: meaning as vectors learned from text, culminating in the composition problem – vector representations lose exactly the structural information (who did what to whom) that the logical tradition captures naturally.

The final and largest unit of the course develops a candidate reconciliation: geometric (Clifford) algebra as a substrate for compositional semantics. We recover the forty-year research tradition of structured vector representations (tensor products, holographic representations, vector symbolic architectures), build the geometric algebra itself – blades, grades, the geometric product, contraction, rotors – and then study Functional Geometric Algebra (FGA), a recent framework in which grade encodes arity, the wedge product implements binding, contraction implements function application, and rotors implement type coercion. The unit opens with the correspondence question directly: how do model-theoretic artifacts map onto vectorial encodings? Students reimplement the course's reference fragment in the geometric substrate and evaluate the comparison first-hand. The course closes with open research problems at this frontier; final projects may engage these directly.

Prerequisites: COSI 10A or COSI 12b or equivalent (introductory programming, any language), and LING 130a (semantics) or COSI 121b (functional programming), or permission of the instructor. **No functional programming experience is required or assumed**; a fully worked functional implementation of the course fragment is provided and studied, not written. Programming assignments use Python. Comfort with basic linear algebra (vectors, inner products, matrices) is strongly recommended for the second half; a self-assessment will be distributed in Week 1, with supplementary materials for review. If you do not meet the criteria but are interested, please talk to us.

Learning goals. Upon completion, students will be able to:

- represent and compute meaning in both logical and distributional frameworks, and articulate the strengths and limits of each;
- carry out type-driven semantic derivations, including quantification and scope, and evaluate them against models;
- read and analyze a working functional implementation of a compositional fragment;
- work with the basic constructions of geometric algebra and apply them to semantic representation;
- carry out and evaluate mappings between model-theoretic artifacts and vectorial encodings;
- read, evaluate, and communicate about contemporary research in computational semantics, including work whose questions are still open;
- integrate these skills in a final project.

Meeting times: Tuesdays and Fridays, 11:10 AM – 12:30 PM. Four credit hours; expect a minimum of nine hours per week of preparation outside class.

Reading Key

- **EU**: van Eijck & Unger (2010), *Computational Semantics with Functional Programming*, selected chapters (Moodle) – read for the semantics; the provided fragment is the executable companion
- **JM**: Jurafsky & Martin (2022+), *Speech and Language Processing*, 3rd ed. draft (online)
- **LS**: Lenci & Sahlgren (2023), *Distributional Semantics*, Cambridge UP (Moodle)
- **D**: Dorst, Fontijne & Mann (2007), *Geometric Algebra for Computer Science* (Moodle)
- **M**: Macdonald (2010), *Linear and Geometric Algebra* – optional, gentler entry for linear algebra review
- **W**: Widdows (2004), *Geometry and Meaning*, CSLI
- **P**: Papers packet (Moodle): Smolensky (1990); Plate (1995); Coecke, Sadrzadeh & Clark (2010); Pustejovsky, *Toward a Functional Geometric Algebra for Natural Language Semantics* (arXiv); FGA lambda notes Parts I–II; Pustejovsky & Krishnaswamy, FPO paper (arXiv); one Clifford neural network paper (TBD)
- **SEP**: Stanford Encyclopedia of Philosophy, "Generalized Quantifiers" (optional, Session 5)

Schedule of Topics

Part I — Meaning as Logic (6 sessions)

Wk	Date	Sess.	Topic	Readings	Assignments
1	Fri 8/28	1	Course introduction; the lambda calculus	—	Linear algebra self-assessment out
2	Tue 9/1	2	Semantic types; type-driven interpretation	EU Ch. 2; JM App. F	
2	Fri 9/4	3	Predicate logic and models	EU 3.6–3.9	HW 1 out
3	Tue 9/8	4	The Fragment of English: lexicon, composition, model checking (EU's notation taught here, as reading knowledge; the provided fragment introduced)	EU Ch. 6	
3	Fri 9/11	5	Quantifiers and scope: generalized quantifiers, ambiguity, readings	EU 7.1–7.4; SEP (opt.)	
4	Tue 9/15	6	Beyond the sentence: modality, intensionality, dynamics (survey)	Course slides	Quiz 1 out (take-home, Part I)

Part II — Meaning as Distribution (4 sessions)

Wk	Date	Sess.	Topic	Readings	Assignments
4	Fri 9/18	7	The distributional hypothesis; co-occurrence, PPMI	LS 2.2–2.4	HW 1 due; Quiz 1 due Mon 9/21
5	Tue 9/22	8	Dimensionality reduction; word2vec; static embeddings	LS 2.4–2.7, Ch. 4	HW 2 out
5	Fri 9/25	9	Contextual embeddings; transformers as semantic models	LS Ch. 5	
6	Tue 9/29	10	The composition problem; GL and word embeddings	LS Ch. 6; GL slides	Project guidelines out

Part III — The Algebraic Thread and Geometric Algebra (5 sessions + flex)

Wk	Date	Sess.	Topic	Readings	Assignments
6	Fri 10/2	11	The algebraic thread in NLP: tensor products, HRR, VSA, DisCoCat	P: Smolensky; Plate; Coecke et al.; W Chs. 1–3	
7	Tue 10/6	12	Geometric algebra I: vectors, bivectors, the geometric product	D Chs. 1–2; M as needed	
7	Fri 10/9	13	Geometric algebra II: multivectors, grades, blades	D Chs. 3–5	HW 2 due
8	Tue 10/13	14	Geometric algebra III: contraction, duality, projection	D Chs. 6–7	
8	Fri 10/16	15	Geometric algebra IV: rotors, signatures Cl(p,q), the assembled toolkit	D Ch. 10	Quiz 2 out (take-home, Parts II–III)
9	Tue 10/20	16	**Flex/consolidation:** worked examples across the GA toolkit; recovery slot if the block ran long	—	Quiz 2 due Mon 10/19

Part IV — Functional Geometric Algebra (8 sessions)

Wk	Date	Sess.	Topic	Readings	Assignments
9	Fri 10/23	17	The correspondence: model-theoretic artifacts as vectorial encodings (entities, predicates, truth values, functions – what maps to what, and what is at stake)	P: FGA paper §§1–2	HW 3 out
10	Tue 10/27	18	FGA I: grade as arity; wedge as binding	P: FGA §3	
10	Fri 10/30	19	FGA II: contraction as unbinding and function application	P: FGA §4; lambda note Pt. I	
11	Tue 11/3	20	FGA III: predication and quantification (the lambda tradition's options – QR, storage, CPS – in one slide; FGA's alternative in depth)	P: lambda note Pt. II	Project proposals due Fri 11/6
11	Fri 11/6	21	FGA IV: coercion as rotor action; type-shifting; GL connections	P: FGA §5; GL readings	

Wk	Date	Sess.	Topic	Readings	Assignments
12	Tue 11/10	22	FGA over knowledge graphs: hyper-relational representation	P: TBD draft/handout	
12	Fri 11/13	23	Learning in FGA; FPO, friction, and alignment	P: FPO paper	
13	Tue 11/17	24	Open problems and the research frontier; project tracks launched in depth	Open-problems handout	HW 3 due Fri 11/20

Projects

Wk	Date	Sess.	Topic	Readings	Assignments
13	Fri 11/20	25	Project workshop: proposals presented, feedback	—	
14	Tue 11/24	—	*No class – Thanksgiving week (confirm Brandeis calendar)*		
15	Tue 12/1	26	Final project presentations I	—	
15	Fri 12/4	27	Final project presentations II	—	Final writeups due during exam period

Dates assume a Fri 8/28 start and standard Brandeis fall calendar; adjust for the actual 2026 academic calendar, the October break/Thursday-schedule day, and reading period. The flex session (16) is the built-in contingency; if the calendar takes a further session, merge Sessions 20 + 21.

Assignments and Assessment

Problem Sets (45%)

HW 1 – Meaning as Logic (15%). Out Week 2, due Week 4. A written problem set: lambda reduction and normal forms; typed derivations; fragment derivations against a model, including quantified NPs and one scope ambiguity with both readings derived; and a guided analysis of the **provided Haskell fragment** (trace specified derivations through the code; predict and verify what the type checker rejects). *Optional alternative for students who want the programming experience:* implement the fragment yourself from the provided scaffold, same weight, same due date – talk to us by the end of Week 2.

HW 2 – Meaning as Distribution (15%). Out Week 5, due Week 7. Build distributional representations from a provided corpus (PPMI and learned embeddings); evaluate on similarity tasks. **Required diagnostic section:** attempt to compose representations for a supplied set of adjective–noun and noun–noun pairs using addition, multiplication, and concatenation; demonstrate and analyze where each fails. The writeup must state precisely *what* information is lost. Python.

HW 3 – The Fragment, Reimplemented in FGA (15%). Out Week 9, due Week 13. Reimplement the course's reference fragment in geometric algebra, in **Python** (clifford / kingdon; scaffold provided): entities and predicates as multivectors, binding via wedge, application via contraction, at least one coercion via rotor. **Required comparison writeup (~3 pp.):** what each substrate makes easy, what each makes awkward, and at least one phenomenon the FGA implementation does not yet handle, with a precise statement of why.

Quizzes (20%)

Both take-home, due the following Monday. **Quiz 1 (10%):** Part I – lambda calculus, types, models, quantification and scope. **Quiz 2 (10%):** Parts II–III – distributional semantics, the composition problem, geometric algebra. Quiz 2 closes before the FGA unit begins so the algebra is consolidated first.

Final Project (30%)

Proposal due Week 11; workshop presentation Week 13; final presentations Week 15; writeup (6–8 pp., ACL format) due during exam period. Seeded **open-problem tracks** are distributed with the project guidelines (Week 6) and developed in depth in Session 24, including: typing structure for graded representations; confluence of contraction-based derivation; copying and reuse (reflexives, anaphora) in exterior-algebra representations; graded objectives for representation learning; friction-based alternatives to standard preference optimization; FGA representations for hyper-relational knowledge graphs. Projects on these tracks are evaluated on the quality of the investigation, not on whether the approach succeeds.

Participation and Labs (5%)

Lab attendance and exercise completion.

Lab Schedule (parallel track)

All programming lives here and in the Python assignments. One lab session – "read, run, break" – carries the executable-fragment experience formerly delivered by the Haskell homework.

- **Weeks 1–2:** Logic workout: reduction, typing, derivation practice; support for HW 1
- **Week 3:** Read, run, break: the provided Haskell fragment – run derivations, feed it ill-typed compositions, watch the type checker refuse; comprehension exercises
- **Weeks 4–6:** Python: corpus processing; embedding construction; support for HW 2
- **Weeks 7–9:** Geometric algebra computation: clifford and kingdon; hand-verification of lecture computations; the 2^n wall and grade truncation
- **Weeks 10–12:** HW 3 support: the Python GA scaffold; multivector debugging; project consultations
- **Weeks 13–15:** Project consultations

Notes to Self (not for publication)

- **The spine principle:** Part I contains exactly what Parts III–IV consume – reduction, types, models, the Fragment, scope-as-phenomenon – plus one survey session (6) for honesty about what lies beyond. Nothing in Part I exists for its own sake.
- **New decks needed:** Session 5 (generalized quantifiers and scope, phenomenon-first, no CPS) and Session 6 (modality/intensionality/dynamics survey) are new lectures. Session 4 folds EU-notation-reading into the Fragment lecture.
- **The provided fragment is now critical-path course material**, doing triple duty: Session 4 exhibit, Lab 3 subject, HW 3 reference. It must be complete, tested, and *pedagogically commented* before the semester starts – it is the Haskell curriculum now. Budget August time.
- **Deck renumbering from Rev 2:** algebraic thread = Session 11; GA I–IV = Sessions 12–15 (the old combined GA III deck still needs its split executed); correspondence = Session 17.
- **Session 20 (FGA III) now carries the scope comparison** – one slide on QR/storage/CPS as the lambda tradition's options, then FGA's treatment. Students know the phenomenon from Session 5 but not the mechanisms; the slide is informational, not a callback.
- **Session 16 flex discipline:** if the GA block lands on time, run prepared worked examples (or an extra hour on signatures); do not backfill with new required content, or the contingency evaporates.
- **Session 12 is no longer the hinge – Session 10 is** (composition failure). Sessions 12–15 remain the risk point; the flex session is the structural mitigation.
- **Identity shift is real:** the course is now "computational semantics: from logic to geometry." Update cs-135.org front page, catalog copy, and any advising language. COSI 121b: "helpful but not required."
- **Enrollment:** this is the widest funnel of the three revisions – no FP requirement, Python throughout, LING-friendly HW 1. Watch whether the strongest MS students take the optional Haskell alternative; if none do, drop it in a future year.
- Assign the external Clifford-NN paper alongside the FGA papers; at least one HW 3 writeup prompt and one quiz question should ask what FGA does not yet handle.
- Open-problems handout for Session 24: distinguish established results, work in progress, and open questions explicitly.